MILITARY SPECIFICATION

CONTROL, RECEIVER-TRANSMITTER C8728( )/VVG-1: 11738851

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electro-mechanical unit known as Control, Receiver-Transmitter C-8728( )/VVG-1: 11738851 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedures for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester
F11738851	Control, Receiver-Transmitter: C-8728( )/VVG-1: 11738851

PACKAGING DATA SHEET

11738851	Packaging of Control, Receiver-Transmitter C-8728( )/VVG-1: 11738851

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Receiver-Transmitter Control C-8728( )/VVG-1, herein referred to as the unit, shall be manufactured in accordance with Drawing F11738851 and drawings pertaining thereto.

3.2 General specification. The unit shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Electrical and electronic assemblies

e. Part identification and marking

f. Workmanship.

3.2.1 Soldering. Soldering of electrical connections shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the unit shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Power and signals. The unit shall perform as specified herein when the power and signals of table I are applied as specified.

3.3.1.1 Continuity. Direct continuity (resistance equal to or less than 0.3 ohm) shall exist between the pins of J1 and P1 which are directly interconnected as shown on the applicable schematic diagram called out on Drawing F11738851.

3.3.1.2 RANGE and RESET switches. With no power or signals applied, the circuit conditions between pins of P1 and between pins of J1, as listed in table II, shall be as specified in table II for the indicated positions of the RANGE and RESET switches.

3.3.1.3 RESET indicator. With the +13 Vdc power of table I applied as specified, but with no signal applied, the indicator light marked RESET shall light.

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TABLE I
POWER AND SIGNALS

Item	Conditions	Characteristics	Connections
1	Power source		Connected between the following pins, when specified herein.
1.1	+13 Vdc	Tolerance: ±1.0Vdc Ripple: ≤25 mV p-p	J1-AA(+) and J1-L(-) J1-M(+) and J1-L(-) J1-J (+) and J1-L (-)
2.	Signal source		Applied as specified herein.
2.1	Digital Signal	Logical one: +13.0 ±1.0V Logical zero: +0.5 ±0.5V	

TABLE II
RANGE AND RESET SWITCHES CIRCUIT CONDITIONS

Item	Switch	Switch Position	Pins of J1		Pins of P1
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1	RANGE	Normal	O	NA	0
2		Depressed	S	NA	S
3	RESET	Normal	NA	O	NA
4		Depressed	NA	S	NA

NOTE 1. Circuit condition legend:
S = Short Circuit (resistance equal to or less than 0.3 ohm)
O = Open Circuit (resistance greater than 1 megohm)
NA = Not applicable

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3.3.1.4 RANGE indicator. With the +13 Vdc power of table I applied as specified, but with no signal applied, the indicator light marked RANGE shall not light. With the +13 Vdc power applied as specified and with the logical zero signal type B of table I applied to J1-N, the range indicator light shall light.

3.3.1.5 Reticle light control. With the +13 Vdc power of table I applied as specified, full-range clockwise rotation of the shaft of potentiometer R1 shall cause the voltage at P1-10 referenced to J1-L to vary from +0.5 ±0.5 Vdc to +13.0 ±1.5 Vdc.

3.3.1.6 Azimuth drive train backlash. With the azimuth output shaft locked in place, the azimuth control knob shall rotate not more than 10 degrees (sum of positive and negative rotation) when a torque of 15 ounce-inch (oz.-in.) is alternately applied in the positive and negative directions.

3.3.1.7 Elevation drive train backlash. With the elevation output shaft locked in place, the elevation control knob shall rotate not more than 8 degrees (sum of positive and negative rotation) when a torque of 15 oz.-in. is alternately applied in the positive and negative directions.

3.3.2 Environmental.

3.3.2.1 Storage temperature. The unit shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperatures. The unit shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3.1 shall be met at ambient temperature.

3.3.2.3 Vibration. The unit shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a.	Amplitude:	1/64 inch, constant (1/32 inch double amplitude)
b.	Vibratory motion:	Simple harmonic
c.	Sweep time:	5 to 55 to 5 Hertz in a five-minute period

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d. Vibration cycling: Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

3.3.2.4 Shock. The unit shall meet the requirements of 3.3.1 after being exposed to a total of 15 shock impulses, three in each direction along the X and Y axes and one in the positive direction along the Z axis. Each shock impulse shall be half-sine wave with a time duration of 3.5 ±0.5 milliseconds and a peak amplitude of 100g.

3.3.3 Sealing. With the unit initially pressurized with nitrogen or air to 5 ±1 pounds per square inch gage (psig) (with cover in place), the loss of pressure shall not exceed 0,055 psig in one-half hour, except for special sampling-environmental, during which the loss of pressure shall not exceed 0.22 psig in 2 hours.

3.3.4 Internal pressure. Subsequent to meeting the sealing requirements, the unit shall be dry flushed with an internal atmosphere containing air or nitrogen having a dew point of at least -63.5°F, and an internal pressure of 3.5 ±0.5 psig shall be established.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specification and Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 First article testing. First article testing shall be conducted in accordance with tables III through V to determine compliance with 4.2. All examinations and tests shall be witnessed by the Government Quality Assurance Representative. When manufacturing processes, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability of the unit, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.3.2.3 Functional tests. The requirements and tests in table III shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

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TABLE III
FUNCTIONAL TESTS

Classification	Characteristics	Requirements	Test Paragraph
101.	Continuity	3.3.1.1	4.6.2
102.	RANGE and RESET switches	3.3.1.2	4.6.3
103.	RESET indicator	3.3.1.3	4.6.4
104.	RANGE indicator	3.3.1.4	4.6.5
105.	Reticle light control	3.3.1.5	4.6.6
106.	Azimuth drive train backlash	3.3.1.6	4.6.7
107.	Elevation drive train backlash	3.3.1.7	4.6.8
108.	Sealing (1/2 hour)	3.3.3	4.6.10

TABLE IV
SPECIAL SAMPLING – GENERAL

Classification	Control Tests	Requirements	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

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TABLE V
SPECIAL SAMPLING — ENVIRONMENTAL

Classification	Control Tests	Requirements	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.9.1
304.	Operating temperature	3.3.2.2	4.6.9.1
305.	Vibration	3.3.2.3	4.6.9.2
306.	Shock	3.3.2.4	4.6.9.3
307.	Sealing (2 hours)*	3.3.3	4.5.10
308.	Internal pressure*	3.3.4	4.6.11
*To be performed after all environmental testing.			

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One unit shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table IV.

4.4.2 Environmental. Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table V.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table VI shall be utilized in performing the tests specified herein.

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TABLE VI
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC Power Supply	+13.0 ±0.7 Vdc Ripple: 25 mV p-p maximum
2.	Ohmmeter	Range: Zero to 10 megohms Accuracy: ±5 percent
3.	Signal Source	Capable of providing dc levels of amplitude specified in item 2.1 of table I.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPs forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

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4.5.2.2.1 Pressure tester. Special test equipment (STE) in accordance with Drawing F8565556, or equivalent, shall be provided to check the sealing of the unit.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.2 Continuity test. Perform resistance measurements between the pins of J1 and P1 which are directly interconnected, to determine compliance with 3.3.1.1.

4.6.3 RANGE and RESET switches test. Perform resistance measurements between the pins of J1 and between the pins of P1, as specified in table II, for each position of the RANGE and RESET switches to determine compliance with 3.3.1.2.

4.6.4 RESET indicator test. Apply the +13 Vdc power as specified in table I, and observe the lighting of the RESET indicator lamp to determine compliance with 3.3.1.3.

4.6.5 RANGE indicator test. Apply the +13 Vdc power as specified in table I, and then apply the logical zero signal type B of table I to J1-N. Observe the RANGE indicator lamp before and after application of the logical signal to determine compliance with 3.3.1.4.

4.6.6 Reticle light control test. Apply the +13 Vdc power as specified in table I. Measure the output voltage at P1-10 while rotating the shaft of potentiometer R1 full-range clockwise to determine compliance with 3.3.1.5.

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4.6.7 Azimuth drive train backlash test. Lock the azimuth output shaft in place. Apply 15 oz.-in. torque in the positive then negative directions to determine compliance with 3.3.1.6.

4.6.8 Elevation drive train backlash test. Lock the elevation output shaft in place. Apply 15 oz.-in. torque in the positive and negative directions to determine compliance with 3.3.1.7.

4.6.9 Environmental tests.

4.6.9.1 Temperature test. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the unit may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.2 through 4.6.8 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.9.2 Vibration test. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part I of Procedure I therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the unit for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.2 through 4.6.8 to determine compliance with 3.3.2.3.

4.6.9.3 Shock test. Perform this test utilizing apparatus specified in Method 516.1 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.2.4, to determine compliance with 3.3.2.4.

4.6.10 Sealing test. Perform this test utilizing the STE specified in 4.5.2.2.1. Pressurize the unit as specified in 3.3.3. Measure the pressure loss over the applicable time period to determine compliance with 3.3.3.

4.6.11 Internal pressure test. Perform this test utilizing the STE specified in 4.5.2.2.1. Verify that the requirements of 3.3.4 are met.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11738851. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Receiver-Transmitter Control C-8728( )/VVG-1 is a unit of the Fire Control Rangefinder AN/VVG-1( ). The unit provides an interlock function for various signals and control functions for the:

a. Range signal

b. Manual reset signal

c. Reticle light

d. Range and manual reset indicator lights

e. Azimuth and elevation mechanical drive trains.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging and packing

d. Applicable stock number

e. Requirement for First Article Submission.

Custodian:

Army - MU

Preparing activity:

Army - MU

PROJECT NO. 1240 - A310

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